TECHNICAL RESEARCH DRAFT

NETRA-4D

Privacy-Aware Temporal Evidence for Examination Surveillance

A candidate–workstation contextual pipeline for per-frame inference, sustained behavioral evidence, real-time delivery, and human adjudication.

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5 VIDEOS	**15,346** FRAMES	**19** EVENT CANDIDATE TRACKS
29 REVIEW EVENTS	**9** EVIDENCE STILLS	**856.8s** SOURCE DURATION

Abstract and Research Scope

Abstract

NETRA-4D is a privacy-aware computer-vision pipeline for transforming examination video into time-aligned review evidence. It combines person pose, object detection, ByteTrack identities, face masking, a 2-D head-direction gaze proxy, candidate-to-workstation association, and temporal persistence. The system emits annotated video, exact source timestamps, structured JSONL event records, high-confidence snapshots, clips, and real-time browser telemetry. It is designed to prioritize auditable human review rather than issue autonomous cheating verdicts.

Research objective

Evaluate whether a modular temporal and contextual reasoning layer can organize real per-frame detections into useful examination-review evidence while preserving identity continuity, source-time semantics, and explicit uncertainty.

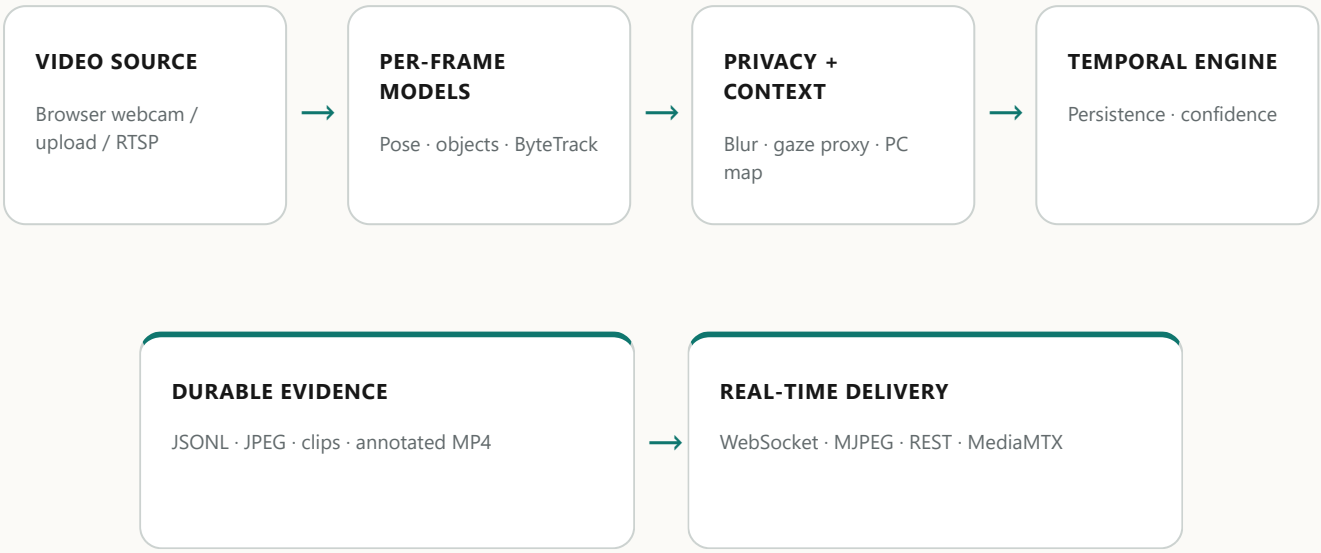
Primary contributions

(1) Lossless frame-by-frame processing for uploaded evidence, with low-latency stale-frame handling for live sources. (2) Candidate and assigned-PC mapping with workstation-relative attention and interaction proxies. (3) Confidence accumulation tied to sustained behavior rather than a single frame. (4) Privacy masking before display and persistence. (5) A complete backend-to-frontend path using REST, WebSocket telemetry, MJPEG delivery, RTSP/MediaMTX ingest, SQLite review state, and verified proof-of-work fallback media.

Evaluation boundary

The run reports measured counts and runtime provenance. It does not report precision, recall, F1, or cheating-detection accuracy because no adjudicated ground-truth corpus was available. Event confidence is an evidence score for prioritization, not the probability that misconduct occurred.

System Architecture and Data Flow



Input policy

Uploaded files use decoder back-pressure and source timestamps (frame index / FPS), ensuring every decoded frame is processed. Webcam and RTSP feeds discard stale queued frames when inference falls behind, preserving current-view latency.

Single-source data contract

Every live session exposes one coherent source: annotated frames, current detections, candidate and workstation assignments, temporal events, evidence references, counters, and health state all share the same session identifier. The frontend therefore cannot accidentally mix telemetry from one source with imagery from another.

Output roots

OUTPUT/live_sessions stores active source state; OUTPUT/live_runs stores completed live evidence; OUTPUT/netra4d_final_20260823 stores the validated offline proof-of-work run. Browser-compatible H.264 samples provide a deterministic fallback when live inference or device permissions are unavailable.

Methodology

Person pose and identity

YOLO26 pose generates person boxes and 17 COCO keypoints. ByteTrack provides candidate identity continuity. Candidate IDs are evidence-track identifiers, not biometric identities.

Objects and small-item review

YOLO26 object detections retain their raw class and confidence. Cell-phone detections remain review evidence. Low-hand possible-chit events are behavioral proxies until a genuinely labeled custom chit model is trained and validated.

Privacy and gaze

Facial keypoints define a face/head mask applied before rendering or persistence. Gaze is a 2-D head-direction proxy from eye/nose geometry; it is explicitly not iris tracking.

Candidate–PC context

Tracked candidates are associated one-to-one with workstation screens where geometry supports the mapping. Assigned device labels use PC-<candidate ID>. Unassigned chairs are omitted; assigned-chair overlays are limited to the initial configured period.

Temporal confidence

Evidence begins with model or geometry signals and increases only while the behavior remains active. The sustained boost is bounded, and activation/crossing timestamps are written using source-video time.

Human review

Events are persisted with lifecycle, evidence kind, score, source signals, candidate/workstation IDs, exact timestamps, keyframes, clips, and specialist-resolution fields. Final disposition belongs to a reviewer.

Measured Results from the Validated GPU Run

19
EVENT-INVOLVED CANDIDATE TRACK IDs

10
CANDIDATE TRACK IDs MAPPED TO PCs

4
TEMPORAL EVENT TYPES

177.6s
SUMMED EVENT PERSISTENCE

6.13s
MEAN EVENT DURATION

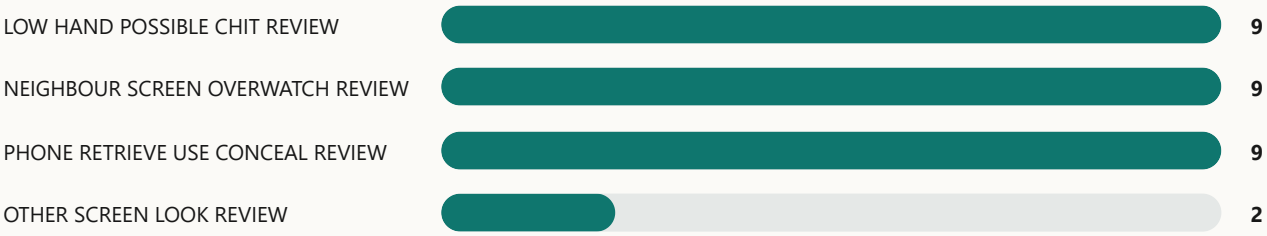
143.75s
MAX EVENT DURATION

39,481
PRIVACY MASK OPERATIONS

12,337
KNOWN GAZE-PROXY OBSERVATIONS

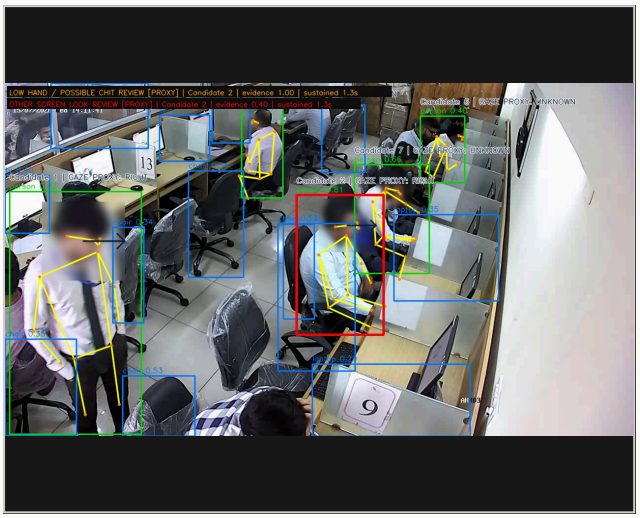
Asset	Frames	Duration (s)	Event candidate IDs	Event types	Events	Evidence stills
VIDEO_01	2,210	88.4	10	4	13	4
VIDEO_02	1,145	143.1	3	3	7	3
VIDEO_03	3,282	131.3	5	2	6	1
VIDEO_04	5,304	212.2	0	0	0	0
VIDEO_05	3,405	281.8	1	2	3	1

Temporal event distribution

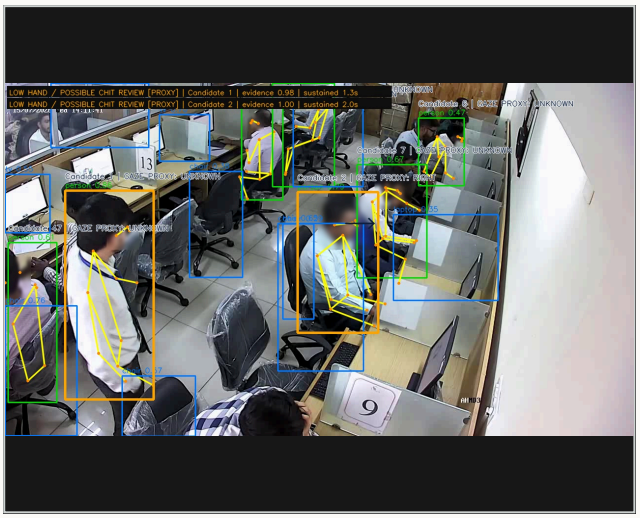


Temporal coverage: source-local event timestamps range from 1.25s to 151.78s; exact start, crossing, duration, confidence, and candidate IDs remain in each JSONL record. **Candidate counts are session-local track IDs, not biometric person counts**, because tracker IDs can fragment after long occlusions.
Runtime provenance: Tesla P100-PCIE-16GB · CUDA device cuda:0 · Torch 2.7.1+cu118 · Python 3.12.13. Frame accounting and hashes are preserved in `run.json` and `COMPLETE.json`.

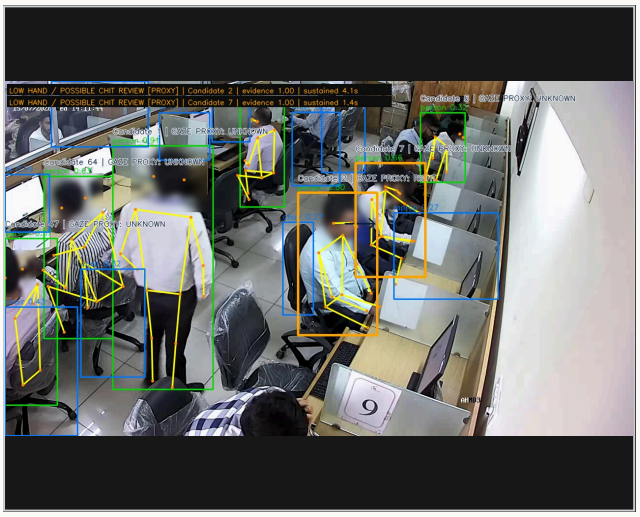
High-Confidence Evidence Gallery



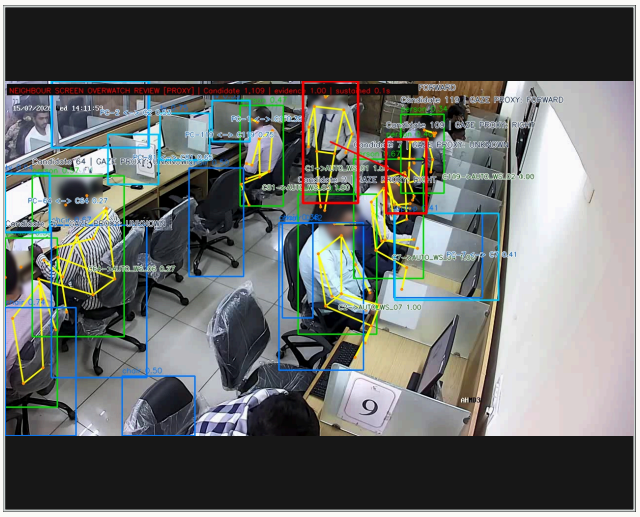
VIDEO_01 · frame 000054 00 t 2.160 conf 1.000 low hand possible chit review



VIDEO_01 · frame 000073 00 t 2.920 conf 0.976 low hand possible chit review



VIDEO_01 · frame 000125 00 t 5.000 conf 1.000 low hand possible chit review



VIDEO_01 · frame 000466 00 t 18.640 conf 1.000 neighbour screen overwatch review

Selection rule. A snapshot is written when a sustained review event crosses the configured evidence threshold. Filename fields retain frame index, exact source timestamp, evidence confidence, and event label. Images are privacy-masked and intended for reviewer inspection.

Limitations, Ethics, and Conclusion

Known limitations

No reviewed ground-truth labels are available, so accuracy metrics are intentionally absent. The validated run produced no raw cell-phone still at the strict COCO confidence threshold of 0.70. A trained custom chit checkpoint is not present. Head direction is not eye gaze. Automatic workstation calibration is camera-dependent and may fail under severe occlusion or insufficient stable observations. CPU live inference can run below source FPS; GPU deployment is recommended for higher throughput.

Risk controls

Face/head masking precedes display; exact timestamps and provenance are retained; all behavioral labels include review semantics; uncertainty and counter-evidence remain visible; local file integrity gates exclude incomplete or size-mismatched runs; and no event is converted into a disciplinary conclusion without a human decision.

Conclusion

The validated evidence demonstrates a coherent source-to-interface pipeline: real videos are decoded, inferred, tracked, privacy-masked, contextually mapped, temporally accumulated, persisted, and streamed to a browser. Scientific claims about misconduct classification require a separately labeled dataset, blinded human adjudication, predefined metrics, and external validation.

References

1. Y. Zhang et al., “ByteTrack: Multi-Object Tracking by Associating Every Detection Box,” ECCV, 2022.
2. Ultralytics, official YOLO model and tracking documentation; checkpoint hashes are recorded in the run manifest.
3. OpenCV, video decoding, encoding, and image-processing runtime used by the evidence pipeline.
4. NETRA-4D internal artifacts: `run.json`, `detections.jsonl`, `review_events.jsonl`, `COMPLETE.json`, and CBT workstation maps.